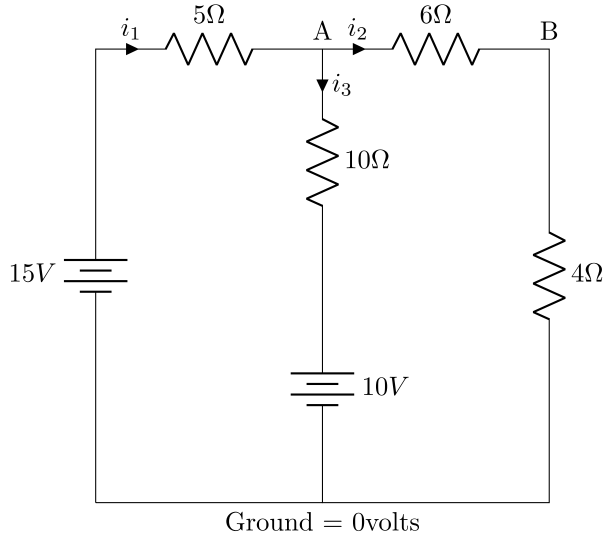
AELL101 Lecture-5

Divyesh Agarwal

Tuesday 21stJanuary, 2025

1 KCL and KVL Practice Questions/Answers:

1.1 Question: In the circuit given below, find the value of current i_1, i_2, and i_3.



Applying KCL at Node A:

$$i_1 = i_2 + i_3$$

1. **Equation:**
$$\frac{15 - v_A}{5} = \frac{v_A - v_B}{6} + \frac{v_A - 10}{10}$$

Observation: The 6Ω and 4Ω resistors are in series.

Conclusion: The current passing through them is the same.

2. $i_2 = i_2$, thus:

$$\frac{v_A - v_B}{6} = \frac{v_B - 0}{4}$$

Simplifying:

$$4v_A - 4v_B = 6v_B$$

$$4v_A = 10v_B$$

$$2v_A = 5v_B$$

Solving the equations:

$$v_A = 10\,\mathrm{V}, \quad v_B = 4\,\mathrm{V}$$

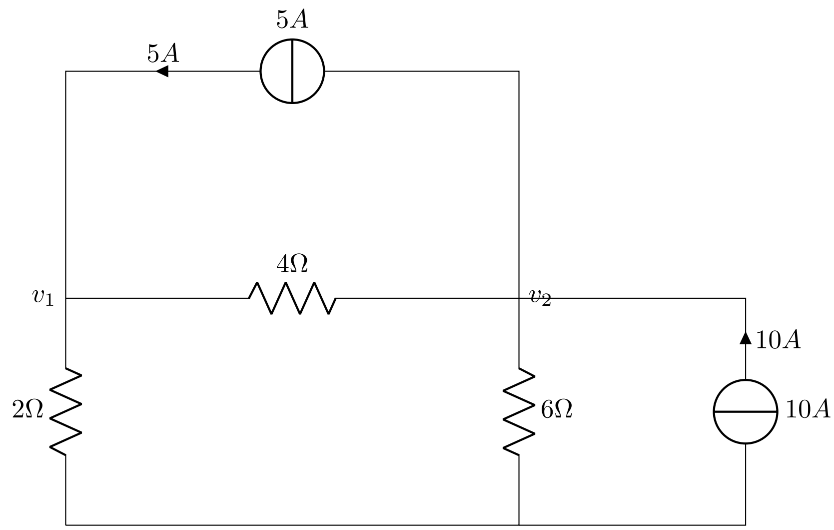
3. **Current Calculations:**

$$i_1 = \frac{15 - 10}{5} = 1\,\mathrm{A}$$

$$i_2 = \frac{10 - 4}{6} = 1\,\mathrm{A}$$

$$i_3 = \frac{10 - 10}{10} = 0\,\mathrm{A}$$

1.2 Question: In the circuit given below, find the values of voltages v_1 and v_2.



Applying KCL at v_1 and v_2 nodes: Assume $v_1 > v_2$.

1. **For v_1:**

$$5 = \frac{v_1 - v_2}{4} + \frac{v_1}{2}$$

Simplifying:

$$20 = v_1 - v_2 + 2v_1$$

$$20 = 3v_1 - v_2$$

2. **For v_2:**

$$10 + \frac{v_1 - v_2}{4} = 5 + \frac{v_2}{6}$$

Simplifying:

$$120 + 3(v_1 - v_2) = 60 + 2v_2$$

$$60 = 5v_2 - 3v_1$$

3. **Solving the equations:** From the first equation:

$$20 + v_2 = 3v_1$$

From the second equation:

$$3v_1 = 5v_2 - 60$$

4. Substituting $3v_1$ from the first equation:

$$20 + v_2 = 5v_2 - 60$$

Simplifying:

$$80 = 4v_2$$

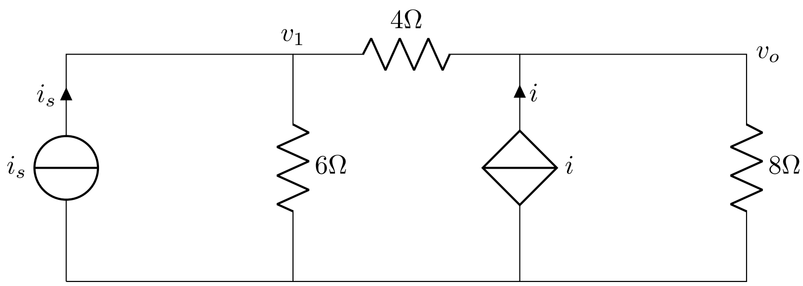
$$v_2 = 20\,\mathrm{V}$$

5. Substituting v_2 back into $3v_1 = 20 + v_2$:

$$3v_1 = 20 + 20$$

$$v_1 = \frac{40}{3}\,\mathrm{V}$$

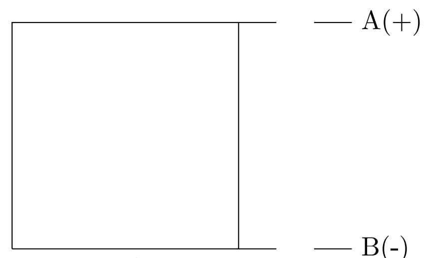
1.3 Question: In the circuit given below, find the value of current i_s. (Assume the value of voltage v_o = 16volts and for the dependent current source, $i = v_1/4$)



1. As $v_0 = 16\,\mathrm{V}$ across the 8Ω resistor, the current in that branch is given as:
$$\frac{v_0}{R} = \frac{16}{8} = 2\,\mathrm{A}$$
2. Assuming $v_1 > 16\,\mathrm{V}$, **applying KCL**:
$$\frac{v_1 - 16}{4} + i = 2$$
Simplifying:
$$2v_1 = 24$$
$$v_1 = 12\,\mathrm{V}$$
Thus, the direction is reversed.
3. **Calculating** i_S:
$$i_S + \frac{16 - 12}{4} = \frac{12 - 0}{6}$$
$$i_S + 1 = 2$$
$$i_S = 2 - 1 = 1\,\mathrm{A}$$

2 Thevenin's Theorem:

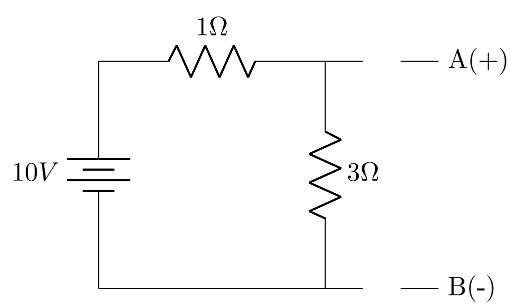
According to this theorem, the whole circuit can be simplified to a smaller one having a equivalent circuit resistance in series to a voltage source.



Fixed part of the circuit

Find v_{AB} across open ends A and B, and then short-circuit to find i_{IS} in AB branch. (Thevenin Voltage = v_{AB} and Thevenin Resistance $r_{TH} = v_{AB}/i_{IS}$)

2.1 Example: Finding the value of v_{TH}, i_{SC}, and r_{TH}.



1. The Thevenin Voltage v_{TH} is given as:

$$v_{TH} = v_{AB} = 7.5\,\mathrm{V}$$

2. The short-circuit current i_{SC} is:

$$i_{SC} = \frac{10}{1} = 10\,\mathrm{A}$$

   Since no current is flowing through 3Ω resistor when A and B are short-circuited.

3. The Thevenin Resistance r_{TH} is calculated as:

$$r_{TH} = \frac{v_{AB}}{i_{SC}} = \frac{7.5}{10} = 0.75\Omega$$

ELL101 Lecture 6

Aryan Jha

January 2025

1 Thevenin Analysis

Example 1. In the given circuit, find the values of the voltages V_1, V_2 and V_3. Solution 1. There are two methods to solve this problem, using loop equations or nodal analysis.

Loop Equations: Assume that 3 A flows in the loop in the bottom right, i_x A in the upper and $2i_x$ A in the bottom left of the circuit shown in Figure 1.1. Also, each current is considered to have clockwise direction. So, loop equation for upper loop,

$-4i - 8(i - 2i_x) - 2(i - 3) = 0$

Also, in the 2 Ω resistance, the current is i_x towards the left, so,

$i_x = 3 - i$

Solving the equations simultaneously, we have,

$i = 1.8A$ and $i_x = 1.2A$

Thus, $V_1 = 4.8V$ and $V_2 = 2.4V$ and $V_3 = -2.4V$

Nodal Analysis: The nodes labelled as point A, B, and C can be analysed to yield three equations in 4 variables. Also, relation between i_x and voltages can be found using Ohm's law,

$\frac{V_2}{4} + \frac{V_2-V_1}{2} + \frac{V_2-V_3}{8} = 0$

$\frac{V_1-V_2}{2} - 3 + \frac{V_1-V_3}{4} = 0$

$2i_x + \frac{V_2-V_3}{8} + \frac{V_1-V_3}{4} = 0$

$i_x = \frac{V_2-V_1}{2}$

Solving the equations simultaneously yields $i_x = 1.2A$, $V_1 = 2.4V$, $V_2 = 4.8V$ and $V_3 = -2.4V$

Example 2. In the given circuit (Fig 1.2), find the Thevenin equivalent circuit along AB.

Solution 2. The first step in finding the equivalent Thevenin circuit is to find the voltage across AB when it is opened, i.e, there exists no 2 Ω resistor in the circuit. Using KVL, we have current in the loop $i = 4.2A$ Thus, voltage across AB $V_{AB} = 11.2V$. This is called the Thevenin voltage V_{th}. Now, we short the AB wire such that no resistor exists in AB wire and find the current through AB. Again using KVL, $i_{sc} = 14A$.

Thus, the Thevenin resistance is $R_{th} = \frac{V_{th}}{i_{sc}} = \frac{11.2}{14} = 0.8\Omega$

Thus the equivalent Thevenin circuit is as shown in Fig 1.3

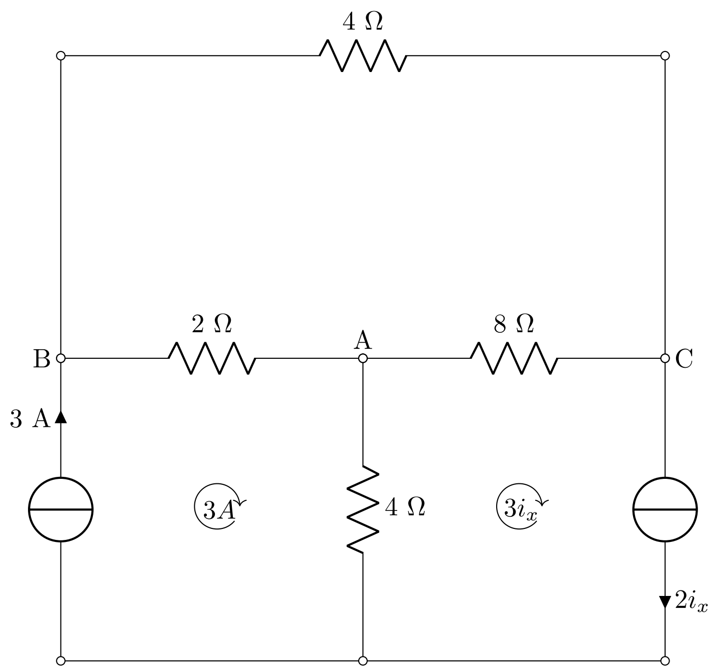


Figure 1: Circuital diagram

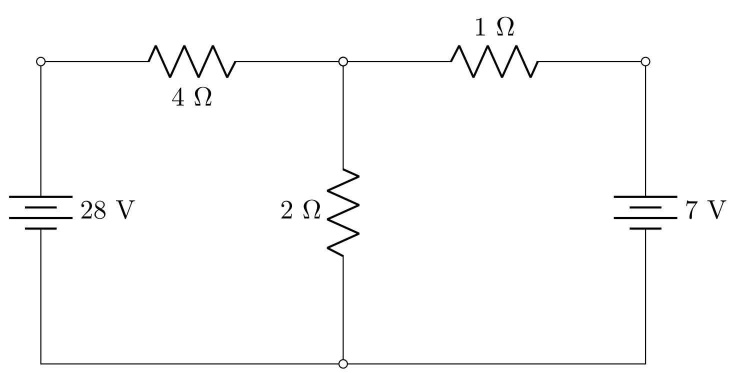


Figure 2: Circuital Diagram

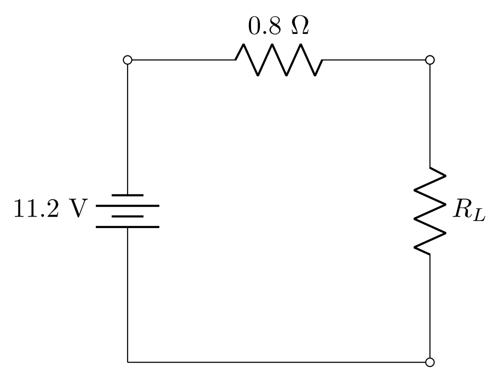


Figure 3: Thevenin Transformation of circuit 2

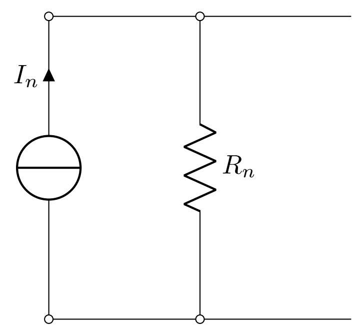


Figure 4: Norton's template circuit

2 Norton Analysis

Just as Thevenin Analysis converts a circuit into a voltage plus unified resistance in series, Norton's analysis helps transform a complex circuit into a current source plus unified resistance in parallel. Both the transformations are quite similar in theory, in that we must find the open circuit voltage across the points chosen (V_{oc}) and then the short circuited current (i_{sc}) across those points. Thus,

$R_n = \frac{V_{oc}}{i_{sc}} I_n = i_{sc}$

It is noteworthy to mention that any Thevenin's circuit can be converted to a corresponding unique Norton's circuit.

AELL101 Lecture-7

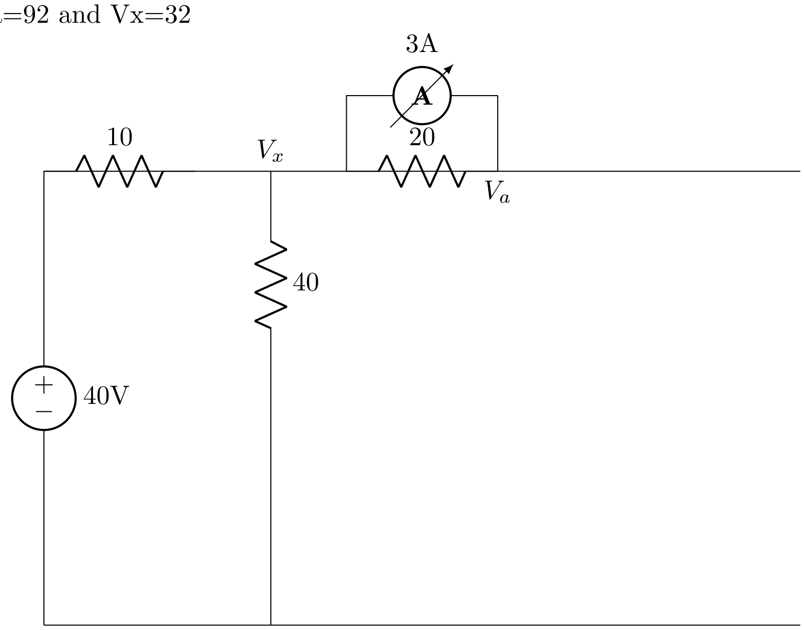
Mohsin Akram Khan, Lawmsanga Renthlei

January 2025

1 Problems

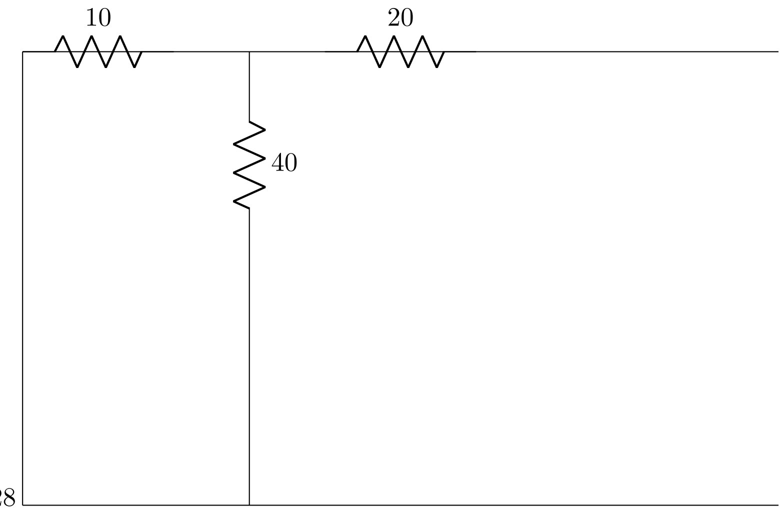
1.1 Find Voltage Vx and Va

(Vx-40)/10 + Vx/40 + 3 + (Vx-Va)/20 = 0
at a : (Va-Vx)/20=3
using these two equations we get
Va=92 and Vx=32



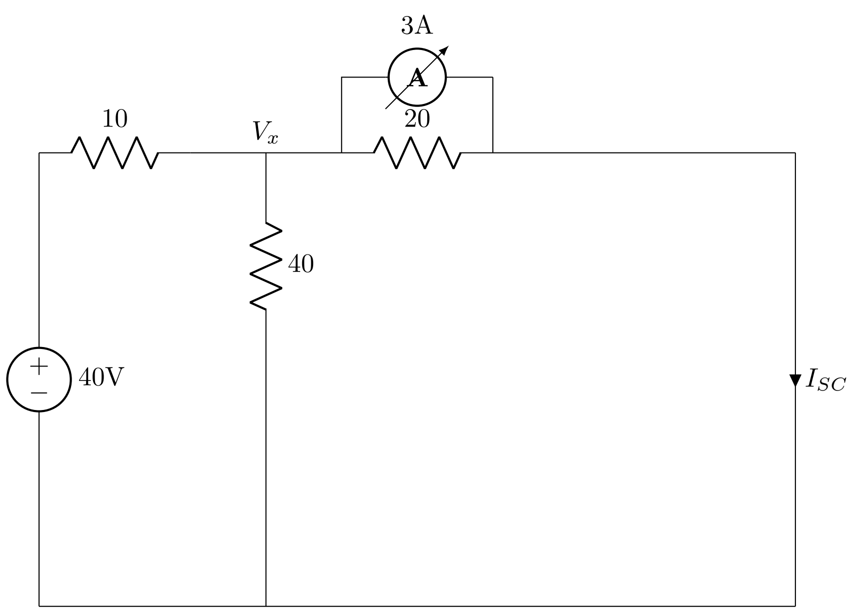
1.2 Rth by dirrect method:

turn off the sources



Here, Req=28

2 Normal method without Rth:



Calculate Isc

Applying Node method assuming potential at unknown node to be Vx we get:

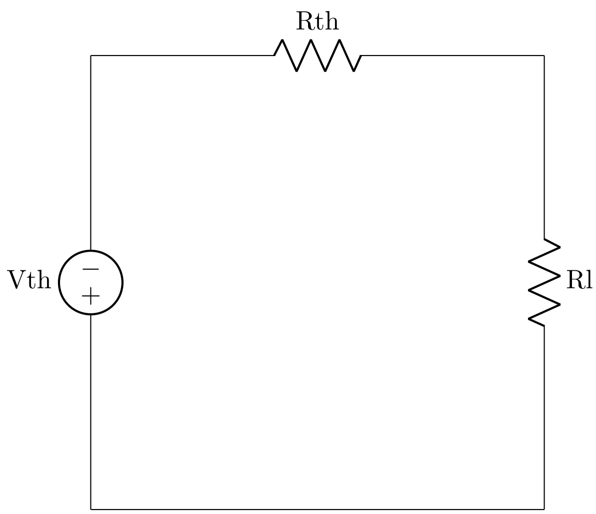
(Vx - 40)/10 + Vx/40 + 3 (Vx - 0)/20 = 0

(4Vx - 160 + Vx + 120 + 2Vx) / 40 = 0

solving above equation, we get Vx=40/7 and Isc = 3 + Vx/20=3.28A

> Load is any equipment where we draw power. There is no load which is purely resistive it will always have some inductance/capacitance

2.1 Find Power across R(load)

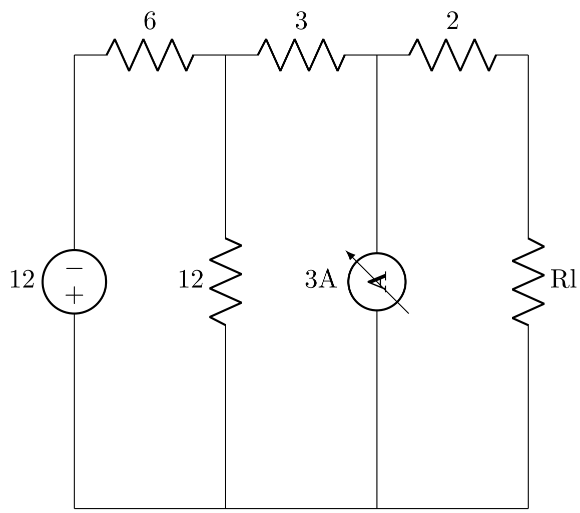


Powerby Rl =Vr*Vr/Rl=i*i*Rl
if Rl=0 then Power at Rl =0(Vr=0: No volatge across Rl)
if Rl tends to infinity then Power at Rl =0(as there is no current flowing)

3 Maximum Power Transfer:

Maximum power is transfer to the load from a network when load resistance equals the thevanin resistance seen from the load
i=Vth/(Rth+Rl): Pl=i*i*Rl=Vth*Vth/(Rth+Rl)*(Rth+Rl)*Rl
Power ismax when Rth=Rl
Pmax=Vth*Vth/(4*Rl)

3.1 Find Rl for maximum power transfer



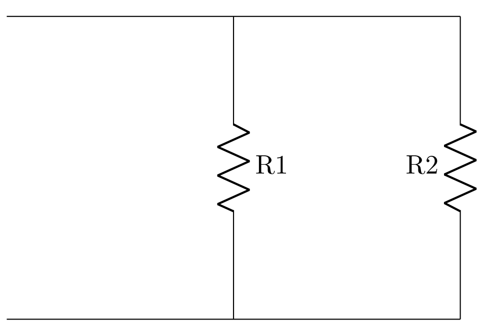
Rnet across Rl=9

Rth direct method does not work when we have dependent voltage/current sources in the circuit

4 Resistors in series

here Vr1=V*R1/(R1+R2) Vr2=V*R2/(R1+R2)

5 Resistor in parallel



Here, a large resistance gives less current I1=I*R2/(R1+R2) I2=I*R1/(R1+R2)
it is used as a current divider

AELL101 Lecture-[5 to 7]

Aryan, Lawma, Divyesh, Mohsin

(21st 22nd 23rd)January, 2025

1 Contribution Statement:

1.1 Divyesh Agarwal - Lecture-5

1.2 Aryan Kumar Jha - Lecture-6

1.3 Mohsin Akram Khan and R Lawmsanga - Lecture-7